



Ask Your Neighbor

Consider the following Cobb-Douglas Production Function:

$$Y = zK^{\alpha}N^{\beta}$$

where

$$\alpha + \beta = 1$$

$$\alpha, \beta \geq 0$$

$$\frac{\partial Y}{\partial N} = \beta \frac{Y}{N} = MPN > 0$$

$$\frac{\partial \beta z K^{\alpha} N^{\beta-1}}{\partial N} =$$

$$=$$

$$\beta - 1 < 0 \quad \left\{ (\beta - 1) \beta \frac{Y}{N^2} \right.$$

If N increases, what will happen to GDP (Y)?

- A. The marginal productivity of N (MPN) will be positive and rise as N rises
- ☒ B. The marginal productivity of N (MPN) will stay positive but fall as N rises
- C. The marginal productivity of N (MPN) will be negative and get more negative as N rises
- D. The marginal productivity of N (MPN) will stay negative but get less negative as N rises



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Consider the following Cobb-Douglas Production Function:

$$Y = zK^{\alpha}N^{\beta}$$

where

$$\alpha + \beta = 1$$

$$\alpha, \beta \geq 0$$

Then, all other things being equal, we know that

A. $3 \cdot zK^{\alpha}N^{\beta} > z(3K)^{\alpha}(3N)^{\beta}$

B. $3 \cdot zK^{\alpha}N^{\beta} < z(3K)^{\alpha}(3N)^{\beta}$

C. $3 \cdot zK^{\alpha}N^{\beta} = z(3K)^{\alpha}(3N)^{\beta}$

D. Any of the above

Decreasing Returns to Scale
Increasing Returns to Scale
CRS